

Aflevering 6

Se på det indre produkt

$$\left\langle \begin{pmatrix} a \\ b \\ c \end{pmatrix}, \begin{pmatrix} d \\ e \\ f \end{pmatrix} \right\rangle = ad + 4be + 3cf$$

På $\mathbb{R}^3$

Der oplyses at $\mathcal{V} = (v_1, v_2, v_3)$ med

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, v_2 = \begin{pmatrix} 5 \\ 1 \\ 1 \end{pmatrix}, v_3 = \begin{pmatrix} 4 \\ -1 \\ 0 \end{pmatrix}$$

Der er en basis for $\mathbb{R}^3$

a) Brug gram Schmidt smidt processen for at finde en ortonormal basis

$$u_1 = \frac{1}{\sqrt{\langle v_1, v_1 \rangle}} * v_1 = \frac{1}{\sqrt{1 * 1 + 4 * 0 * 0 + 1 * 1 * 3}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{4}} * \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \end{pmatrix}$$

$$p_1 = \langle v_2, u_1 \rangle u_1 = \frac{1}{4} v_1 \langle v_2, v_1 \rangle = \frac{1}{4} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} (5 + 0 + 3) = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$$

$$u_2 = \frac{v_2 - p_1}{\sqrt{\langle v_2 - p_1, v_2 - p_1 \rangle}} = \frac{\begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}}{\sqrt{\langle \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix} \rangle}} = \frac{\begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}}{\sqrt{9 + 4 + 3}} = \begin{pmatrix} \frac{3}{4} \\ \frac{1}{4} \\ -\frac{1}{4} \end{pmatrix}$$

$$p_2 = \langle v_3, u_1 \rangle u_1 + \langle v_3, u_2 \rangle u_2$$

$$= \left\langle \begin{pmatrix} 4 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \end{pmatrix} \right\rangle \begin{pmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \end{pmatrix} + \left\langle \begin{pmatrix} 4 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{3}{4} \\ \frac{1}{4} \\ -\frac{1}{4} \end{pmatrix} \right\rangle \begin{pmatrix} \frac{3}{4} \\ \frac{1}{4} \\ -\frac{1}{4} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \frac{1}{4} * \left\langle \begin{pmatrix} 4 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix} \right\rangle \begin{pmatrix} \frac{3}{4} \\ \frac{1}{4} \\ -\frac{1}{4} \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \frac{1}{4} (12 - 4) \begin{pmatrix} \frac{3}{4} \\ \frac{1}{4} \\ -\frac{1}{4} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} \frac{3}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{5}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = \frac{v_2}{2}$$

$$u_3 = \frac{v_3 - p_2}{\sqrt{\langle v_3 - p_2, v_3 - p_2 \rangle}} = \begin{pmatrix} \frac{3}{2} \\ \frac{3}{2} \\ -\frac{1}{2} \end{pmatrix} * \frac{1}{\sqrt{\frac{9}{4} + 9 + \frac{1}{4}}} = \begin{pmatrix} \frac{3}{4 * \sqrt{3}} \\ \frac{3}{4 * \sqrt{3}} \\ \frac{1}{4 * \sqrt{3}} \end{pmatrix}$$

$$u = \left(\begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 0 \\ \frac{1}{2} \end{pmatrix}, \begin{pmatrix} \frac{3}{4} \\ \frac{1}{4} \\ \frac{1}{4} \\ -\frac{1}{4} \end{pmatrix}, \begin{pmatrix} \frac{3}{4 * \sqrt{3}} \\ \frac{3}{4 * \sqrt{3}} \\ -\frac{1}{4 * \sqrt{3}} \\ -\frac{1}{4 * \sqrt{3}} \end{pmatrix} \right)$$

Er en ortonormal basis

b) Bestem den ortogonale projektion af vektoren

$$v = \begin{pmatrix} 2 \\ 4 \\ 2 \end{pmatrix}$$

På underrummet $\text{span}(v_1, v_2)$

Ud fra gram schmidt processen har vi

$$\text{span}(v_1, v_2) = \text{span}(u_1, u_2)$$

Ortogonal mængde da kan vi bruge lemme 10.14

$$\begin{aligned} p &= \frac{\langle v, u_1 \rangle}{\langle u_1, u_1 \rangle} * u_1 + \frac{\langle v, u_2 \rangle}{\langle u_2, u_2 \rangle} u_2 = \frac{2 * \frac{1}{2} + 3 * \frac{1}{2} * 2}{1} * u_1 + \frac{\left(\frac{3}{2} + 4 * 4 * \frac{1}{4} + 3 * -\frac{1}{4} * 2 \right)}{1} u_2 \\ &= 4u_1 + 4u_2 = \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \\ 1 \end{pmatrix} \end{aligned}$$